

Robot Framework — Task 3

1. Experiment Title

Custom Keywords and Libraries Using Robot Framework

2. Problem Statement

Build a custom Robot Framework keyword using Python to perform a specific action, such as calculating the sum of two numbers. Also utilize the **BuiltIn** library to manipulate strings or perform operations within the test cases.

3. Objective

To create and use a **custom Python keyword** in Robot Framework and demonstrate the use of the **BuiltIn library** for performing operations within Robot Framework test cases.

4. Tools, Software and Concepts Used

Tools / Software

- Python
- Robot Framework
- Robot Framework BuiltIn library
- Google Chrome — not required for this experiment
- Visual Studio Code

Concepts

- Custom Python keywords
- Robot Framework libraries
- Python functions
- Calculate Sum keyword
- BuiltIn library
- Variables
- Should Be Equal As Integers
- Set Variable

- Get Length
- Log

5. Source Code / Implementation Steps

File 1: custom_keywords.py

```
def calculate_sum(number1, number2):
    return int(number1) + int(number2)
```

This Python function is imported into Robot Framework and becomes the custom keyword:

Calculate Sum

File 2: custom_keyword_test.robot

```
*** Settings ***
```

```
Library    custom_keywords.py
```

```
Library    BuiltIn
```

```
*** Test Cases ***
```

```
Test Custom Sum Keyword
```

```
    ${result}=    Calculate Sum    10    20
```

```
    Should Be Equal As Integers    ${result}    30
```

```
    Log    Sum of 10 and 20 is ${result}
```

```
Test BuiltIn String Operation
```

```
    ${text}=    Set Variable    Selenium Robot Framework
```

```
    ${length}=    Get Length    ${text}
```

```
    Log    String length is ${length}
```

```
    Should Be Equal As Integers    ${length}    24
```

Implementation Steps

1. Create a Python file containing the calculate_sum() function.

2. Import the Python file as a Robot Framework library.
3. Use the custom Calculate Sum keyword with values 10 and 20.
4. Verify that the returned result is 30.
5. Import the BuiltIn library.
6. Use Set Variable to store a string.
7. Use Get Length to calculate the string length.
8. Verify the result using Should Be Equal As Integers.
9. Use Log to display information during execution.

6. Output with Clearly Labelled Screenshots

Caption:

Figure 3: Successful execution of custom keyword and BuiltIn library tests

7. Result

The custom Python keyword was successfully integrated with Robot Framework, and the Calculate Sum keyword correctly calculated the sum of two numbers. The BuiltIn library test also executed successfully.

8. Observation

The Python function was recognized by Robot Framework as a custom keyword. The calculated sum of 10 and 20 was verified as 30. The BuiltIn library successfully performed the string operation and calculated the length of the specified text.

9. Conclusion

The experiment demonstrated how **Python functions can be converted into custom Robot Framework keywords** and how the **BuiltIn library** can be used for common operations and verification within Robot Framework test cases.

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